BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7471

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If the mass of spring mass system is doubled, then its time period becomes:

(A) $\sqrt{2}$ T
(B) T/2
(C) $\sqrt{T/2}$
(D) $T/\sqrt{2}$

اگر سپرنگ ماس سسٹم کے ماس کو دوگنا کر دیا جائے تو اس کا ٹائم پیریڈ ہوگا:

Q2. The unit of intensity of sound is:

(A) Wm^{-2}
(B) Wm^{-1}
(C) Wm^{2}
(D) Wm

آواز کی انٹینسٹی کا یونٹ ہے:

Q3. Snell's Law is:

(A) $n = \sin \hat{r} / \sin \hat{i}$
(B) $n = \sin \hat{i} / \sin \hat{r}$
(C) $\sin \hat{i} = \sin \hat{r}$
(D) $n = C/V$

سنیل کا قانون ہے:

Q4. The value of k in Coulomb's Law is:

(A) $9\times10^{9}Nm^{2}C^{-2}$ (B) $9\times10^{9}Nm^{-2}C^{2}$

(C) $9\times10^{9}Nm^{2}C^{-2}$ (D) $9\times10^{9}Nm^{2}C^{-2}$

کولمب کے قانون میں k کی قیمت ہے:

Q5. Boolean expression of NOR gate is:

(A) $X = A + B$ (B) $X = A \cdot B$

(C) $X = A\cdot\bar{B}$ (D) $X = A+\bar{B}$

نار گیٹ کی بولین علامت ہے:

Q6. The turn ratio of a transformer is 10. It means:

(A) $N_s = N_p/10$ (B) $I_s = 10\ I_p$

(C) $V_s = V_p/10$ (D) $N_s = 10\ N_p$

ٹرانسفارمر کے ٹرنوں کی نسبت 10 ہو تو:

Q7. Mobile phone sends and receives the message in the form of: _____

(A) Sound waves (B) Longitudinal waves

(C) Radio waves (D) Mechanical waves

سیل فون پیغام کو _____ کی صورت میں بھیجتا اور وصول کرتا ہے۔

(A) آواز کی لہروں **(B)** لانگیٹیوڈنل لہروں

(C) ریڈیو لہروں **(D)** مکینیکل لہروں

Q8. The symbol to denote diode is:

(A) [triangle pointing to bar - diode symbol] (B) [NOT gate symbol]

(C) [AND gate with bubble] (D) [triangle with vertical line]

ڈائیوڈ کو ظاہر کرنے کی علامت ہے:

Q9. Generally an atom is represented by the symbol:

(A) A above X, Z below X [exact arrangement [illegible in source scan]] (B) [arrangement [illegible in source scan]]

(C) [arrangement [illegible in source scan]] (D) [arrangement [illegible in source scan]]

عام طور پر ایٹم کو علامت سے ظاہر کیا جاتا ہے:

Q10. Equivalent resistance of the circuit is: [R_1=2Ω, R_2=3Ω, R_3=6Ω in parallel, V=6V]

(A) 11Ω
(B) 36Ω
(C) 6Ω
(D) 1Ω

سرکٹ کی مساوی مزاحمت ہے:

Q11. The shock absorbers damp the vibrations and convert their energy into _____ of the oil.

(A) Kinetic energy
(B) Potential energy
(C) Heat energy
(D) Sound energy

شاک ابزاربرز وائبریشن کو جذب کر دیتے ہیں اور ان کی انرجی کو _____ میں تبدیل کر دیتے ہیں۔

(A) کائی نیٹک انرجی
(B) پوٹینشل انرجی
(C) ہیٹ انرجی
(D) ساؤنڈ انرجی

Q12. Complete the following nuclear reaction: $^{2}_{1}H + ^{3}_{1}H \rightarrow ..... + ^{1}_{0}n$ + Energy

(A) $^{3}_{1}H$
(B) $^{2}_{1}H$
(C) $^{4}_{2}He$
(D) $^{5}_{2}He$

درج ذیل نیوکلیئر ری ایکشن کو مکمل کیجیے:

Physics (Subjective) Group - I

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7471

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]te Ohm's Law and write its mathematical equation.

(ii) [illegible in source scan]ferentiate between conductors and insulators. Give an example of each.

(iii) [illegible in source scan]ite the equation of Gamma decay for cobalt $^{60}_{27}Co$.

(iv) [illegible in source scan]ine electromagnetic induction.

(v) [illegible in source scan]w the force acting on the armature be affected by increasing the strength of the magnetic field?

(vi) [illegible in source scan]ite two properties of alpha and beta particles.

(vii) [illegible in source scan]mplete the given reaction: $^{1}_{0}n + ^{235}_{92}U \rightarrow ^{141}_{56}Ba$ + [illegible in source scan]

(viii) [illegible in source scan]at is the difference between a generator and electric motor?

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]w the depth of sea can be measured by ultrasound?

(ii) [illegible in source scan]und is a form of wave. Give its reason.

(iii) [illegible in source scan]ine capacitance and write its unit.

(iv) [illegible in source scan]culate the frequency of sound waves of speed $340ms^{-1}$ and wave length 0.5m.

(v) [illegible in source scan]es each capacitor carry equal charge in series combination? Explain briefly.

(vi) [illegible in source scan]ine internet and write its two services.

(vii) [illegible in source scan]ite two advantages of electronic mail.

(viii) [illegible in source scan] electric intensity a vector quantity? What will be its direction?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) [illegible in source scan]tinguish between longitudinal and transverse wave with diagram.

(ii) [illegible in source scan]d the time period of simple pendulum of 1m long on moon if $g_m = g_e/6$

(iii) [illegible in source scan]ine principle axis and focal length.

(iv) [illegible in source scan]ine reflection of light and draw its diagram.

(v) [illegible in source scan]at is difference between real and virtual image?

(vi) [illegible in source scan]aw a symbolic diagram for OR gate and write its truth table.

(vii) [illegible in source scan]ine analogue quantities and give an example.

(viii) [illegible in source scan]at is meant by Boolean Algebra?

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) [illegible in source scan]at is meant by electromagnet? Explain the function of relay with diagram.

Q5(b). (Marks 05) [illegible in source scan] applying potential difference of 10V across a conductor, a current of 1.5A passes through it. [illegible in source scan]w much energy would be obtained from the current in 2 minutes?

Q6(a). (Marks 04) [illegible in source scan]at is meant by noise pollution? Write its effects and methods to minimize it.

Q6(b). (Marks 05) [illegible in source scan] capacitor holds 0.03C charge when fully charged by a 6V battery. How much voltage would be [illegible in source scan]uired for it to hold 2 Coulombs of charge?

Q7(a). (Marks 04) [illegible in source scan]plain with the help of activity that waves are carriers of energy.

Q7(b). (Marks 05) [illegible in source scan] object 4cm high is placed at a distance of 12cm from a convex lens of focal length 8cm. [illegible in source scan]culate the position and size of the image.

GROUP – II

Physics (Objective) Group - II

Time Allowed: 15 Minutes	**Maximum Marks:** 12	**Paper Code:** 7476

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Electric potential and emf:

(A) Are the same terms
(B) Are the different terms
(C) Have different units
(D) Both B and C

الیکٹرک پوٹینشل اور ای ایم ایف:

(A) ایک جیسی مقداریں ہیں
(B) مختلف مقداریں ہیں
(C) ان کے یونٹ مختلف ہیں
(D) B اور C دونوں

Q2. Transformer is used to change the value of:

(A) Power
(B) Charge
(C) Voltage
(D) Energy

ٹرانسفارمر استعمال کیا جاتا ہے قیمت بدلنے کے لیے:

(A) پاور
(B) چارج
(C) وولٹیج
(D) انرجی

Q3. The application used to view web pages is called:

(A) Email
(B) Gmail
(C) Browser
(D) MS Office

ایسا سافٹ ویئر جو ویب پیجز کو دیکھنے میں مدد فراہم کرتا ہے کہلاتا ہے:

(A) ای میل
(B) جی میل
(C) براؤزر
(D) ایم ایس آفس

Q4. Boolean expression of NAND gate is:

(A) $X = A + B$
(B) $X = A \cdot B$
(C) $X = A \cdot \bar{B}$
(D) $X = A + \bar{B}$

نینڈ گیٹ کی بولین علامت ہے:

Q5. The half-life of a certain isotope is one day. What is the quantity of the isotope after two days?

(A) One half
(B) One quarter
(C) 8
(D) None of these

ایک مخصوص آئسوٹوپ کی ہاف لائف ایک دن ہے۔ دو دن گزرنے کے بعد آئسوٹوپ کی مقدار کتنی ہوگی؟

(A) آدھی رہ جائے گی
(B) ایک چوتھائی
(D) ان میں سے کوئی نہیں

Q6. Which is not a radio active element?

(A) Uranium
(B) Polonium
(C) Thorium
(D) Sodium

ریڈیو ایکٹیو عنصر نہیں ہے:

(A) یورینیم
(B) پولونیم
(C) تھوریم
(D) سوڈیم

Q7. The example of longitudinal waves is:

(A) Sound waves
(B) Radio waves
(C) Light waves
(D) Water waves

لانگیٹیوڈنل ویوز کی مثال ہے:

(A) ساؤنڈ ویوز
(B) ریڈیو ویوز
(C) لائٹ ویوز
(D) واٹر ویوز

Q8. The pitch of sound mostly depends upon:

(A) Frequency
(B) Period
(C) Wave length
(D) Amplitude

آواز کی پچ کا زیادہ تر انحصار ہے:

(A) فریکوئنسی پر
(B) پیریڈ پر
(C) ویو لینتھ پر
(D) ایمپلیٹیوڈ پر

Q9. The critical angle of glass is:

(A) 42°
(B) 45°
(C) 90°
(D) 0°

شیشے کا کریٹیکل اینگل ہے:

Q10. The pole of spherical mirror is also called:

(A) Principal axis
(B) Focus
(C) Focal length
(D) Vertex

سفیریکل مرر کا پول بھی کہلاتا ہے:

(A) پرنسپل ایکسز
(B) فوکس
(C) فوکل لینتھ
(D) ورٹیکس

Q11. According to Coulomb's Law, what happens to the attraction of two oppositely charged objects as their distance of separation increases?

(A) Increases
(B) Decreases
(C) Remains unchanged
(D) Cannot be determined

کولمب کے قانون کے مطابق اگر دو مخالف چارجز کے درمیان فاصلہ بڑھا دیا جائے تو ان کے درمیان کشش پر کیا اثر پڑے گا؟

(A) بڑھ جاتی ہے
(B) کم ہو جاتی ہے
(C) کوئی تبدیلی نہیں آتی
(D) معلوم نہیں کیا جا سکتی

Q12. How will be the electric lines of force where electric field is strong?

(A) Apart
(B) From positive to negative
(C) From negative to positive
(D) Closer

الیکٹرک لائنز آف فورس کیسی ہوں گی جہاں الیکٹرک فیلڈ کی شدت زیادہ ہوگی؟

(A) دور دور
(B) مثبت سے منفی
(C) منفی سے مثبت
(D) قریب قریب

Physics (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7476

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) State Faraday's Law of electromagnetic induction.

(ii) Differentiate between motor and generator.

(iii) Write the equation of Gamma decay for cobalt $^{60}_{27}Co$.

(iv) Write the causes of background radiation.

(v) Define fusion reaction and write its equation.

(vi) Differentiate between cell and battery.

(vii) Define electric power and write its unit.

(viii) In order to measure voltage in a circuit voltmeter is always connected in parallel. Why?

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Define pitch and quality of sound.

(ii) Write two uses of ultrasound in the medical field.

(iii) Sounds of how much frequency, a normal human ear can hear?

(iv) State Coulomb's Law. Also write its mathematical form.

(v) Define electric field and electric field intensity.

(vi) Differentiate between information technology and telecommunication.

(vii) How does fax machine work?

(viii) In computer, what is the difference between primary memory and secondary storage devices?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) State Hooke's Law and write its equation.

(ii) What are the necessary conditions for a body to execute simple harmonic motion?

(iii) Define the term mechanical wave and write its types.

(iv) Differentiate concave and convex lens by diagram.

(v) How the reflection of light graphically verified?

(vi) Distinguish between regular and irregular reflection.

(vii) Name the four uses of oscilloscope.

(viii) Define the terms digital and analogue quantities.

Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) What is meant by electric current? Explain with the help of diagram that how battery is a current source?

Q5(b). (Marks 05) Cobalt-60 is radioactive element with half-life of 5.25 years. What fraction of the original sample will be left after 26 years?

Q6(a). (Marks 04) What is noise pollution? Explain its sources, effects. How is it reduced?

Q6(b). (Marks 05) The force of repulsion between two identical positive charges is 0.8N, when the charges are 0.1m apart. Find the value of each charge.

Q7(a). (Marks 04) Define wave motion. Explain with example "waves as carriers of energy".

Q7(b). (Marks 05) An object 30cm tall is located 10.5cm from a concave mirror with focal length 16cm. (a) Where is the image located? (b) How is it high?